

Welcome to AST 111 Lab

TA: Spencer Emmons

Introductions

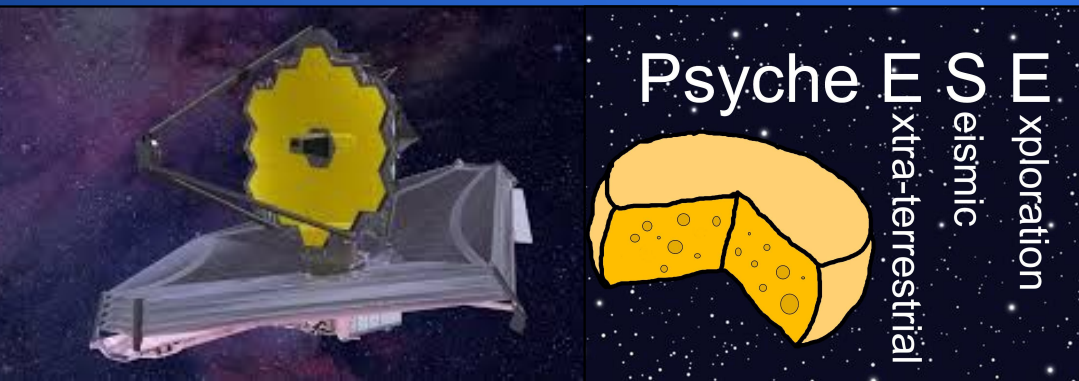
I'm Spencer, I'll be your TA this semester

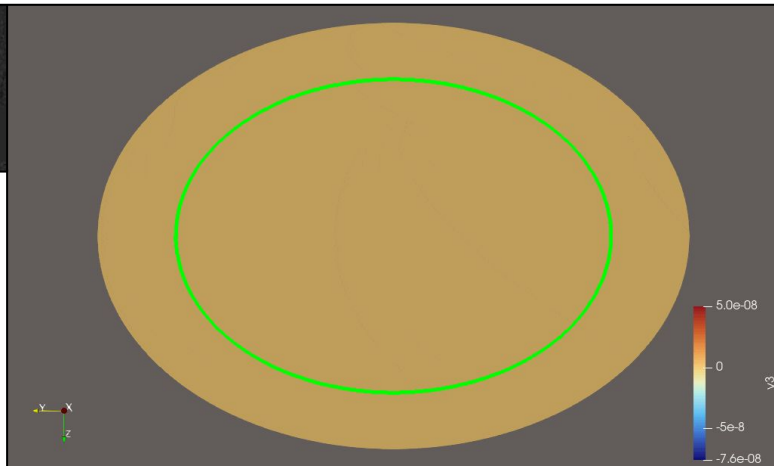
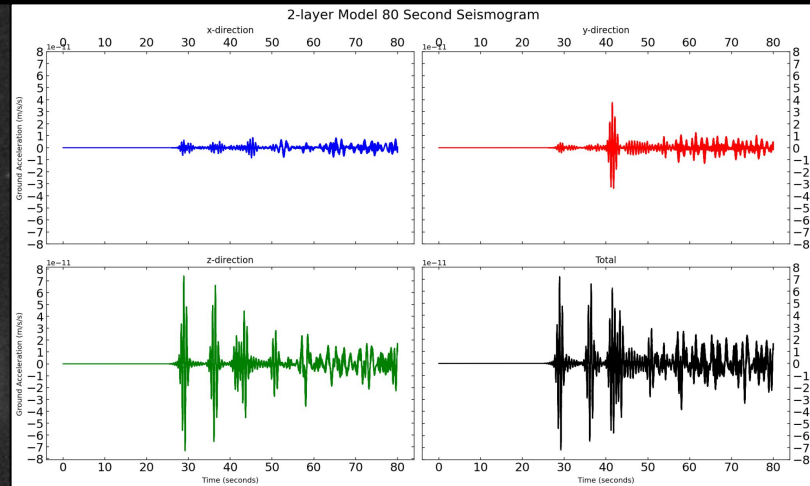
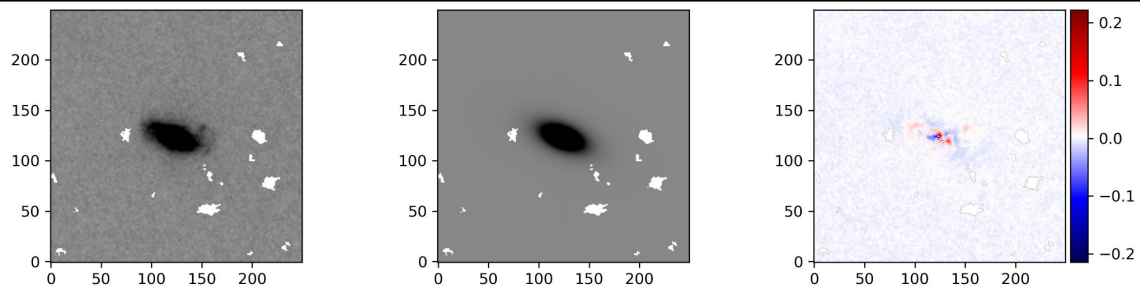
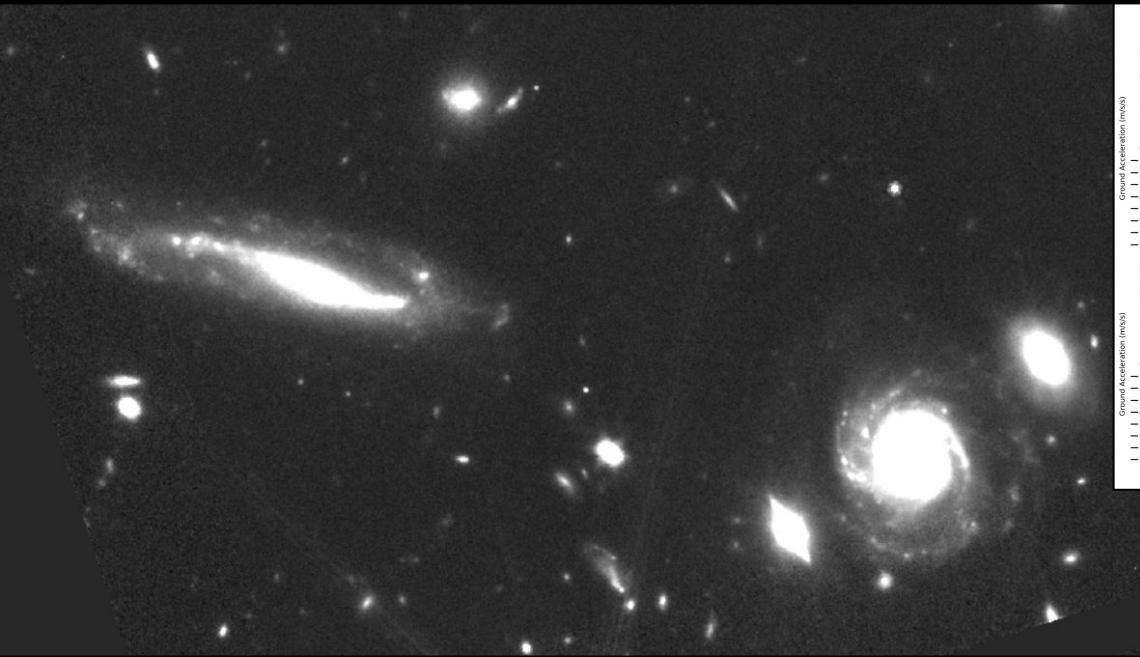
Education

- Associate's in Science from Yavapai College
- Bachelor's in Earth and Space Exploration (Astrophysics) with a minor in Physics and certificate in Planetary Science (major finished, still working on minor and certificate)

Research

- Interests in a broad range of topics in astronomy and related areas
 - I'm particularly interested in planetary science, asteroids, and impact seismology
- Current/past work:
 - Windhorst Cosmology Research Group - Using JWST data to study galaxy evolution and where stars form in those galaxies
 - "Psyche-ESE" - Using computer simulations to understand the seismic effects of man-made impacts on asteroids





<https://semmons98.github.io/research/>

Schedule

- Class will be at **either** 3:30 p.m. **or** 7:30 p.m. each week
 - DSL VR Labs are every other week and need to be done before class, time slots are 4:00 p.m. and 5:00 p.m
- Check the class website: <http://windhorst111lab.asu.edu/> to see the time for each week

Materials

- Star Wheel is **Required**
 - You can buy one at the campus bookstore or print one from the template on the class website
- Scientific Calculator is **Required** (computer/cellphone based is fine)
- A red flashlight or headlamp is **strongly recommended**
 - Has much less impact on the night vision of yourself and others
 - No white/blue lights when we do observations

Communication

- My Office Hours: **1:45-2:45 pm, Wednesdays, in PSF 186**
 - I am open to rescheduling if this does not work for the majority of students
 - Feel free to come by with any questions you have
- Email/Canvas Message
 - semmons3@asu.edu
 - I will typically respond within a couple hours, and always by end of the next school day
 - If you email me after normal business hours (or close to the end of them), you will typically get a response the next morning (exceptions of course when we have night labs like today)
 - Feel free to contact me if you have any questions about the course, the class material, and/or if the current time for office hours does not work for you

Assignments

- DSL VR Experiences need to be completed before class each week
 - Time slots are available at 4:00 p.m. and 5:00 p.m. on the days we have class
- The labs themselves - Due 2 days after we meet - but there should be plenty of time to complete during the lab meetings
 - Two parts, both turned in on Canvas
 - There are 3 exceptions, but I will explain when we get to them
 - In general, you will work in groups but must each submit your own lab
- Quizzes - Due 2 days after each lab
 - Completed on your own - You are allowed your (completed) lab script, and any other materials from the lab or lecture that are relevant - no internet or other people

Grading

- Your transcript will show a single grade for the lecture and lab - weighted $\frac{3}{4}$ for the lecture and $\frac{1}{4}$ for the lab
- Grades are mostly automatic in Canvas, any parts that aren't will typically be graded within a couple days of the due date
- **Always** put an answer down
 - I try to be generous with partial credit, and am willing to go back through quizzes to manually add partial points for fill-in-the-blank questions
- Most of the labs were massively rewritten over the summer
 - So there may be minor mistakes
 - Let me know if you find any errors or think something was graded wrong, I can look into it and get it fixed
- Lowest lab and lowest quiz are automatically dropped

Attendance and Absences

- Attendance is required
 - I will pass a sign-in sheet around each class
 - If you are not present you will receive a 0 for the lab even if you turn it in
- If 3 or more labs are missed (and not excused/made up), you will receive a grade of “E”
 - Regardless of how well you do on other assignments
- Let us know **before** if you are going to miss a lab
 - This way I can figure out a makeup lab for you
 - If there is a genuine emergency, the email can wait

Academic Integrity

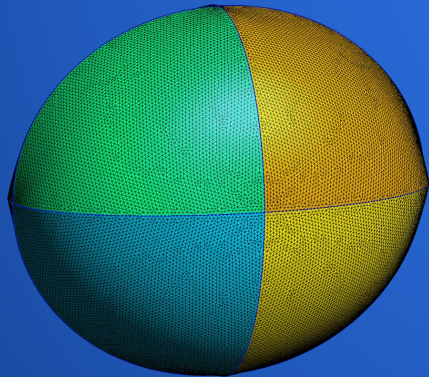
- We take academic integrity seriously
- This includes plagiarism, usage of non-allowed materials (i.e. the internet on quizzes), and any Generative AI usage
 - Generative AI includes Google's AI overview
 - *Almost* all the information needed is in the lab scripts or (sometimes) from the lecture
 - If you need help, my job is literally to provide it, just ask
- Though you are expected, encouraged, and in some cases required to work together, each student must still turn in their own work

LAB 1

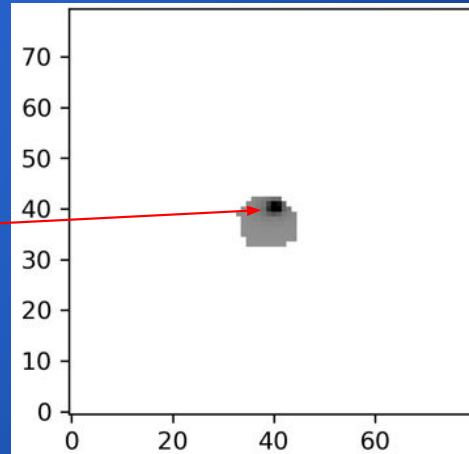
Scientific Models, Artifacts, and Angular Size

Scientific Models

- In science, a model is a way of describing and predicting the behavior of complex systems
 - This can take the form of a set of plain english words that describe what's going on
 - Or a mathematical equation (or a set of numbers)
 - Or a scale model to show something normally too big or small to handle
 - And more!



ellip	0.396
flux	1.976
n	4.682
r_eff	3.596
theta	1.744
xc	40.195
yc	40.402



Artifacts

- In digital images, (and especially astronomical ones) we often see what we call “artifacts”
- These are things in the image that aren’t really there
- Possible causes include:
 - Dead pixels
 - Interference from cosmic rays
 - Dust or damage on the telescope
 - Diffraction spikes
 - Data compression
 - etc.



JWST has a distinctive diffraction spike pattern due to the design of its mirrors and support arms

Today's Lab (Part 1)

- Your first task today is going to be making a model that can identify whether or not something in an astronomical image is an artifact, or an actual object in space
- To do this, work through the first 11 questions, filling out table 1 as you go
- Then questions 12 through 23 will test to see if your model works as a predictive tool
- Step 3 will then be to determine if the object found in the DSL VR experience is an artifact or a real astronomical object

Angular Size

- Things that are further away, appear smaller than they actually are
- Angular size, is a way to measure how large they appear using degrees
 - a.k.a. How much of the sky is this thing taking up
- A useful way to think about this is that if you spin around you can see 360°
 - Angular size is how many of those degrees does an object cover
- If you know 2 out of the angular size, the actual size, and the distance; you can calculate the other 1
 - Unfortunately for us, we only know the angular size of the object in the DSL VR
 - (see question 28)

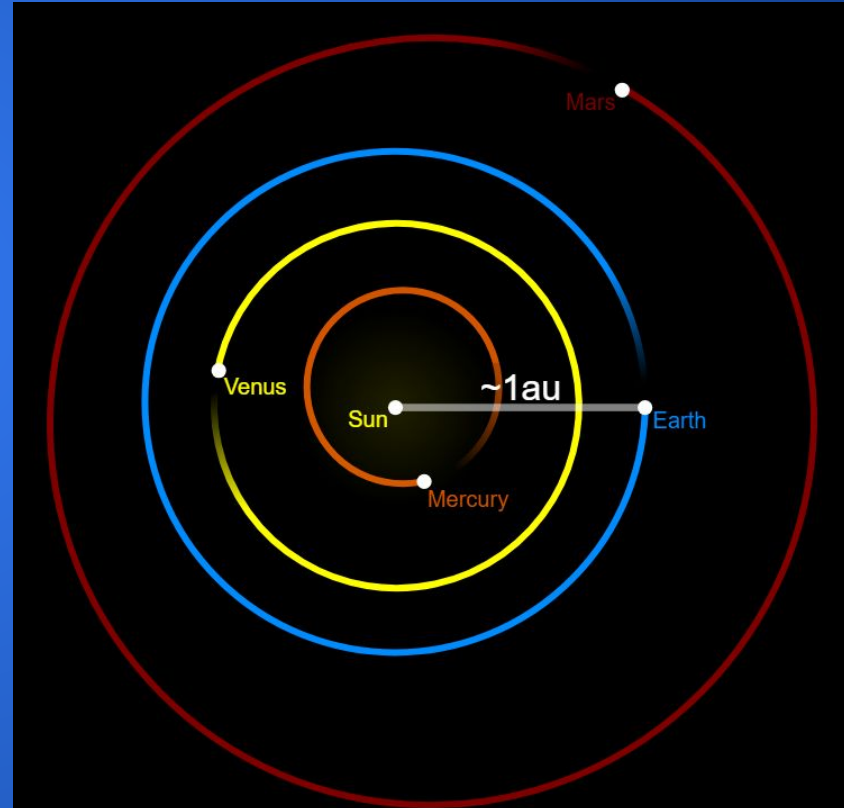


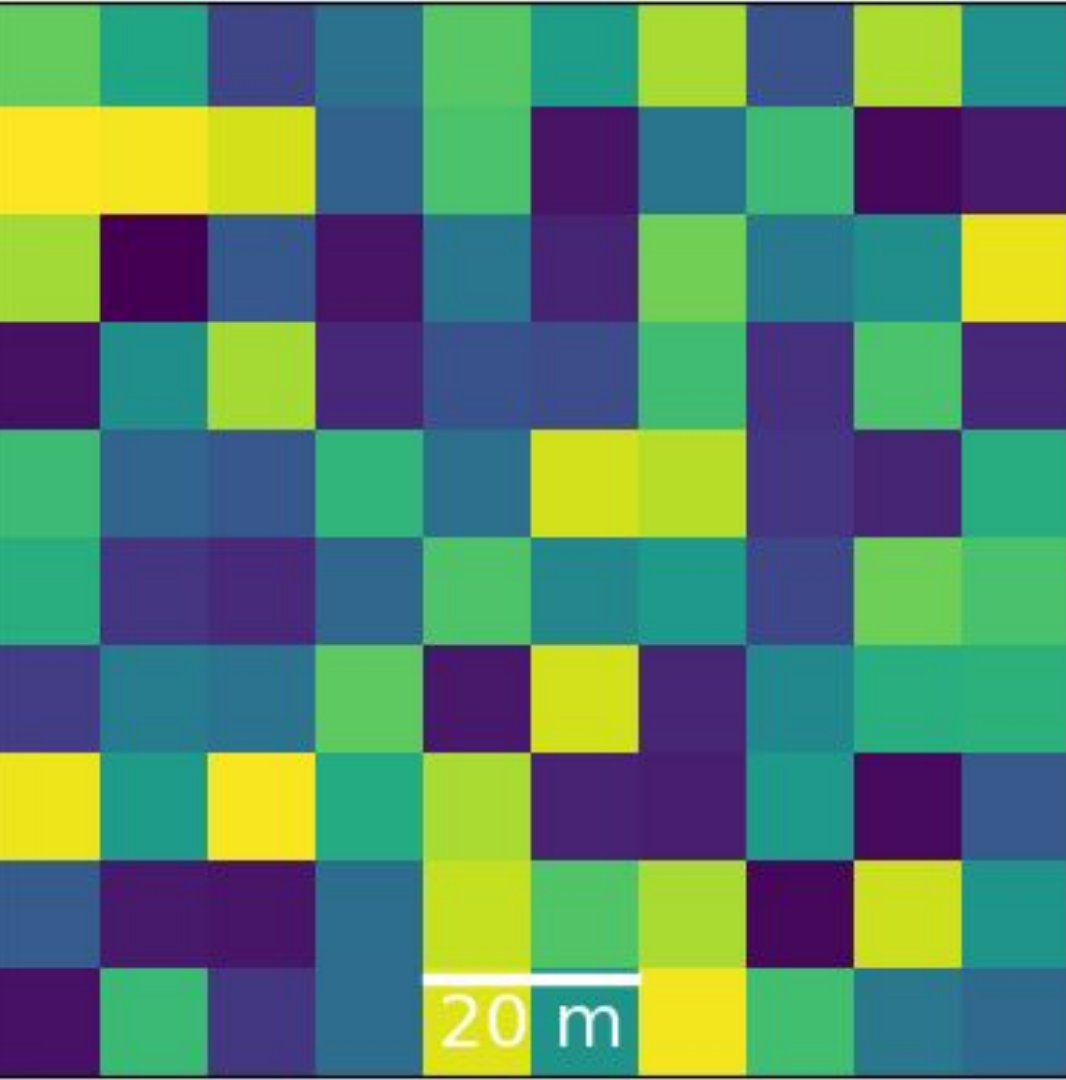
The moon has
an angular size
of about 0.5°

Scaling Things Down (Or Up)

- There are a lot of ways of writing scale when talking about scale models
 - 28mm - meaning the average human is 28mm tall at that scale
 - 1:64 or 1/64 - meaning the scale model is 1/64th of the size of the real thing
 - What you are working with today (for distances) is going to be 1 AU = 2.5 cm - meaning that every 2.5 centimeters of your scale model, represents 1 AU in real life
 - You will also need to understand a scale bar
 - On maps and images a scale bar is a small bar with a number and unit next to it that serves as a size reference for the map or image

AU is “Astronomical Unit”, it is equal to the distance between the Earth and the Sun





Scaling Things Down (Or Up) (Cont.)

← the scale bar in this image says 20 meters, so if something takes up half that scale bar, it is 10 meters and if takes up two scale bars, it is 20m

This is useful to estimate the size of things in images

Today's Lab (Part 2)

- Calculate the scale distances for Table 3 (Distance Model)
 - To do this, multiply the number of AU you are given by 2.5 to get a distance in centimeters
 - Then mark all these distances out on a piece of receipt paper to see how far apart the planets are
- Identify which of the model planets represents which real planet (Physical Model)
 - Use the scale $1 D_{\text{Earth}} = 1.5 \text{ cm}$ ($1 D_{\text{Earth}}$ means you are using the diameter of the Earth as the unit) to fill in table 5
 - Note that 4 of the models do not represent real planets
- Now, you are going to build a model showing the relative angular sizes
 - One person holds Mars in front of themselves at arms' length, then another holds Saturn
 - The person holding Saturn moves back until Mars exactly covers Saturn (with rings), use this for question 33
- The last major step is to fill in Table 6 by calculating the angular size of each planet
 - Use this equation: $\theta = 2.06 \cdot 10^5 \cdot \frac{D}{d}$, where D is the size of the object and d is the distance (this will give an angular size in arcseconds (part of a degree))